Pulsar Timing Arrays: Theory, Instrumentation & GW Physics

Radio Astronomy — March 24, 2026

Name: _____

Date: _____

0 WARM-UP: The PTA as a Receiver System

Foreshadowing: PTAs as Receiver Arrays

In standard communications or sensor engineering, a *receiver array* detects a wave by cross-correlating signals between pairs of receivers. The correlation between two receivers depends only on their geometric arrangement relative to the incoming wave field — this is called the **overlap reduction function (ORF)**.

Jenet & Romano (2015, arXiv:1412.1142) show that *exactly* this structure appears in pulsar timing arrays. A PTA is a receiver array, millisecond pulsars are the transmitters, and the gravitational wave background is the "signal field." The ORF for GR tensor modes takes the specific form of the **Hellings-Downs curve** — a result that is generic to any isotropic receiver/transmitter system in the framework of General Relativity.

Before filling in the analogy table below, briefly consider: what are the key figures of merit (sensitivity, bandwidth, noise, dynamic range, . . .) for any transmitter–receiver system operating at GHz frequencies? Think about what each maps onto for a PTA. Your instructor will draw out the analogy on the board.

Pre-table brainstorm: Key figures of merit for a GHz transmitter–receiver system, and what role they play:

The PTA as a Receiver/Transmitter Analogy

Generic System	PTA Equivalent	Key Point
Transmitter	_____	Stable astronomical clock; not controlled by us
Carrier frequency	_____	Broadband; dispersed by the ISM
Message / modulation	_____	"Message" IS the precision of _____
Channel noise	_____	Must calibrate _____ dispersion
Receiver	_____	$T_{\rm sys}$, $\Delta\nu$ determine _____
Signal processing	_____	Folding, dedispersion, matched filtering
SNR / sensitivity	_____	Timing precision $\rightarrow$ GW sensitivity

Key unusual feature of the PTA system (fill in):

_____

The big idea (to keep in mind all lecture):

The cross-correlation of timing residuals between pulsars a and b depends on the *geometry* of the two Earth-pulsar baselines relative to the GW sky. For an isotropic, unpolarized GW background in GR, averaging over all incoming GW directions gives a universal function of only the *angle between baselines* — the Hellings-Downs curve. This is the ORF of the PTA receiver array.

Different GW polarisation states (GR predicts + and $\times$ tensor modes; alternative theories predict scalar or vector modes) give *different* ORFs. The HD curve is therefore the unique fingerprint of GR gravitational waves.

1 Pulsar Basics and the Timing Model

What is a Millisecond Pulsar (MSP)?

Key MSP Properties

- Spin period: $P \sim$ _____ ms (vs $\sim$s for normal pulsars)
- Spin-down rate: $|\dot{P}| \sim$ _____ (vs 10^{-15} for normal pulsars)
- "Recycled" via _____ from a binary companion
- Clock stability rivals _____; no _____ or glitches

Quick question: Why is a large moment of inertia $I \sim 10^{45}\,\mathrm{g\,cm^2}$ key to MSP clock stability?

The Timing Residual

Fill in blanks and complete derivations as your instructor presents them on the board.

Barycentric Correction

Correct the telescope arrival time $t_{\rm obs}$ to the Solar System Barycenter (SSB):

$$t_{\rm SSB} = t_{\rm obs} + \underbrace{\qquad\qquad}_{\text{Rømer delay}} - \underbrace{\qquad\qquad}_{\text{ISM dispersion}} + \underbrace{\qquad\qquad}_{\text{relativistic corrections}}$$

The **Rømer delay**:

$$\Delta_\odot = _____$$

(define $\hat{\boldsymbol{n}}$ and $\boldsymbol{r}_\oplus$ here: _____)

The **dispersion delay**:

$$\Delta_{\rm DM} = \frac{e^2}{2\pi m_e c} \cdot \frac{\rm DM}{f^2}, \qquad {\rm DM} = \int_0^d n_e\, dl$$

Units of DM: _____. Why does $\Delta_{\rm DM} \propto f^{-2}$? _____

The **relativistic corrections** $\Delta_{\rm rel}$ include: _____

The Timing Model and Residual

The **timing model** predicts pulse arrival times as a power series in time:

$$t_{\text{pred}} = t_0 + \sum_{k=1}^{N} \frac{\nu^{(k)}}{k!}(t - t_0)^k + \cdots$$

This is a Taylor expansion *motivated by* _____.

Here $\nu^{(k)} \equiv d^k\nu/dt^k\big|_{t_0}$ is _____.

The model also includes _____ (for pulsars in binaries).

The **timing residual** is defined as:

$$R(t) = _____$$

If the timing model is perfect: residuals are _____.

Any unmodelled effect produces a _____ in the residuals.

A gravitational wave produces a _____ in the residuals.

Quick question: If a GW passes *only* the Earth (not the pulsars), what is the spatial character of the residuals across all pulsars in the array? Is this correlated or uncorrelated? What pattern would you expect?

2 GW Physics: BBH Inspirals and PTA Sensitivity

GW Strain from a Circular Binary

Polarisation Amplitudes

The **chirp mass**:

$$\mathcal{M} = _____ \quad \text{(in terms of } m_1, m_2\text{)}$$

GW polarisation amplitudes for an inspiraling binary at distance d, inclination angle ι (see derivation in Buonanno 2007, arXiv:0709.4682):

$$h_+(t) = -\frac{4G\mathcal{M}^{5/3}}{c^4 d}(\pi f_{\rm gw})^{2/3} \cdot _____ \cdot \cos(2\pi f_{\rm gw} t)$$

$$h_\times(t) = -\frac{4G\mathcal{M}^{5/3}}{c^4 d}(\pi f_{\rm gw})^{2/3} \cdot _____ \cdot \sin(2\pi f_{\rm gw} t)$$

$f_{\rm gw} = _____ \times f_{\rm orbital}$ (explain why: _____)

Derivation workspace

Derive the sky- and inclination-averaged strain amplitude h_0.

Start from $h_+(t)$ and $h_\times(t)$ above. Average $\langle h_+^2 \rangle$ and $\langle h_\times^2 \rangle$ over inclination angle $\iota \in [0, \pi]$, then average over sky position (solid angle). Show that:

$$h_0 = \frac{4(G\mathcal{M})^{5/3}(\pi f_{\rm gw})^{2/3}}{c^4 d}$$

Hint: $\langle (1+\cos^2\iota)^2/4 \rangle_\iota = _____$, $\langle \cos^2\iota \rangle_\iota = _____$

Peters' Formula: GW-Driven Inspiral

Orbital Evolution Under GW Emission (Peters 1964)

Energy of a circular orbit: $E =$ _____ (gravitational binding energy)

GW luminosity from quadrupole radiation:

$$\frac{dE}{dt} = -\frac{32}{5}\frac{G^4}{c^5}\frac{m_1^2 m_2^2 (m_1 + m_2)}{a^5}$$

Setting $\dot{E}_{\rm orbit} = \dot{E}_{\rm GW}$ and using Kepler's 3rd law ($f_{\rm orb}^2 = GM/4\pi^2 a^3$), derive:

$$\frac{da}{dt} = _____$$

For $\mathcal{M} \sim 10^8 M_\odot$ at $f_{\rm gw} \sim 10^{-8}$ Hz, estimate $\dot{f}/f^2$ over a 15-yr observation.

Is the binary **monochromatic** in the PTA band? _____

Derivation workspace

Workspace: derive and estimate $\tau_{\rm merge}$ for a $10^9 + 10^9\, M_\odot$ BBH at initial orbital separation $a_0 = 0.01$ pc.

Starting from da/dt above, integrate to find $\tau_{\rm merge}$ as a function of a_0, m_1, m_2. Write the formula here, then substitute numbers.

$$\tau_{\rm merge} \approx _____$$

Characteristic Strain and the SGWB Spectrum

Characteristic Strain

The **characteristic strain** of a GW signal:

$$h_c(f) = h_0\sqrt{\frac{f^2}{\dot{f}}}$$

Why this combination? h_c weights the signal by the number of cycles per logarithmic frequency interval, making it the natural quantity for comparison with broadband noise floors.

For a single circular, GW-driven binary, using $\dot{f} \propto \mathcal{M}^{5/3} f^{11/3}$:

$$h_c(f) \propto f \text{ —— } \mathcal{M} \text{ —— } d \text{ ——}$$

Summing over a cosmological population of SMBHBs (following Phinney 2001, arXiv:astro-ph/0108028):

$$h_c^2(f) = \frac{4G}{\pi c^2 f^2} \int dz\, d\mathcal{M}\, \frac{dE_{\rm gw}}{df_r} \frac{d^3n}{dz\, dt_r\, d\mathcal{M}} \frac{1}{(1+z)}$$

For circular, GW-driven inspirals this integral yields the **phenomenological power law**:

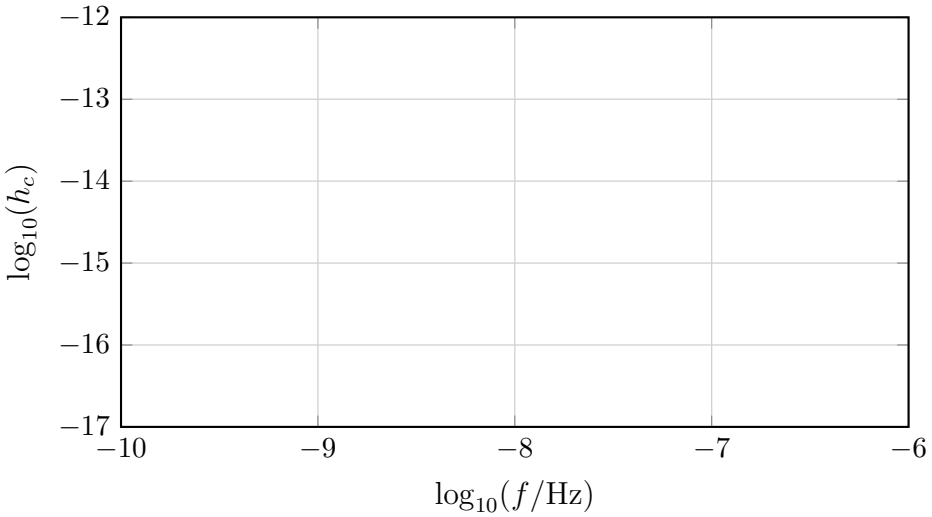
$$\boxed{h_c(f) = A_{\rm gw}\left(\frac{f}{f_{\rm yr}}\right)^{\alpha}}$$

where $f_{\rm yr} = (1\,{\rm yr})^{-1}$, $\alpha =$ _____, and $A_{\rm gw} \approx$ _____ (from NANOGrav 15-yr).

What does a steeper (more negative) spectral index at low f indicate?

Sketch the characteristic strain spectrum on the axes below (add a power-law line with the correct slope; label $A_{\mathbf{gw}}$ and $f_{\mathbf{yr}}$):

Sketch: SGWB characteristic strain (add power-law line with correct slope!)



Individual Sources vs. Stochastic Background:

- A **stochastic GW background (SGWB)** arises from $\sim 10^4$–10^6 **unresolved** BBH systems throughout the universe superposing incoherently.
- Nearby, massive ($\mathcal{M} \gtrsim 10^9 M_\odot$) BBHs may produce a **continuous GW (CGW)** source: a deterministic, quasi-sinusoidal signal resolvable as an individual event.
- Individual **merger events** ("bursts") may be detectable for very massive systems within the PTA band.
- The **Poisson regime** (above $\sim$few nHz): a handful of individually bright sources can exceed the smooth background level.

Write the key distinction:

SGWB → _____ correlation between pulsar pairs;

individual CGW → _____ signal in residuals of each pulsar.

3 The Hellings-Downs Curve: Derivation

Follow the board derivation carefully. This is the mathematical centrepiece of PTA science.

Step 1: GW-Induced Timing Residual

GW Redshift and the Pulsar Antenna Pattern

A GW propagating in direction $\hat{\boldsymbol{k}}$ induces a **fractional redshift** in the pulse frequency of a pulsar in direction $\hat{\boldsymbol{p}}$ (see, e.g., Jenet & Romano 2015):

$$z(t, \hat{\boldsymbol{k}}) = \frac{1}{2} \frac{\hat{p}^i \hat{p}^j}{1 + \hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{p}}} \left[h_{ij}(t_E, \hat{\boldsymbol{k}}) - h_{ij}(t_P, \hat{\boldsymbol{k}}) \right]$$

This z is the fractional change $\delta\nu/\nu$ in pulse frequency; it does *not* depend on the radio carrier frequency ν.

Note: this is a *different quantity* from the timing residual defined in Section 1. Reconcile: the timing residual is the time-integral of this redshift:

$$R(t) = _____$$

(consistent with $R = t_{\text{obs}} - t_{\text{pred}}$ since $z = \delta\nu/\nu = -\delta t_{\text{arrival}}/t$)

The **antenna pattern** (response function) for polarisation $A \in \{+, \times\}$:

$$F^A(\hat{\boldsymbol{k}}, \hat{\boldsymbol{p}}) = \frac{\hat{p}^i \hat{p}^j}{2(1 + \hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{p}})} e^A_{ij}(\hat{\boldsymbol{k}})$$

where e^+_{ij} and $e^\times_{ij}$ are the GW _____ tensors.

Earth Term vs. Pulsar Term

The redshift involves two evaluations of h_{ij}:

- **Earth term**: h_{ij} evaluated at time $t_E = t$ at Earth's location — this term is *coherent* across all pulsars for a given GW.
- **Pulsar term**: h_{ij} evaluated at time $t_P = t - D/c$ at the pulsar's location (where D is the Earth-pulsar distance) — this term differs between pulsars because each pulsar is at a different distance.

Earth term: evaluated at _____, dominates because

_____.

Pulsar term: evaluated at _____, effectively averages to zero in the SGWB correlation because _____.

Step 2: Cross-Correlation and the Overlap Reduction Function

Building the Hellings-Downs Integral

For an **isotropic, Gaussian, unpolarised SGWB** the GW strain is a random field. Its two-point correlation in the frequency domain is:

$$\langle \tilde{h}_A(f, \hat{\boldsymbol{k}}) \tilde{h}^*_{A'}(f', \hat{\boldsymbol{k}}') \rangle = \frac{S_h(f)}{8\pi} \delta_{AA'} \delta^{(2)}(\hat{\boldsymbol{k}}, \hat{\boldsymbol{k}}') \delta(f - f')$$

where $S_h(f)$ is the _____, the $\delta_{AA'}$ enforces _____,

the $\delta^{(2)}$ enforces _____, and the $1/8\pi$ is a _____ convention.

The cross-power of timing residuals from pulsars a and b at frequency f:

$$C_{ab}(f) = \Gamma(\zeta_{ab}) \cdot S_h(f)$$

The **overlap reduction function (ORF)**, $\Gamma(\zeta_{ab})$, is the sky-averaged product of antenna patterns:

$$\Gamma(\zeta_{ab}) = \int \frac{d\hat{\boldsymbol{k}}}{4\pi} \sum_{A \in \{+,\times\}} F^A_a(\hat{\boldsymbol{k}}) F^A_b(\hat{\boldsymbol{k}})$$

(Earth-term only, short-arm approximation; see instructor notes for why the exponential phase factor and pulsar terms drop out.)

Derivation workspace

Set up the coordinate system and evaluate the sky integral.

Coordinate setup: Place pulsar 1 along $\hat{z}$; place pulsar 2 in the xz-plane at angle ζ from pulsar 1, so $\hat{\boldsymbol{p}}_2 = (\sin\zeta, 0, \cos\zeta)$.

GW propagation direction: $\hat{\boldsymbol{k}} = (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta)$.

Step 1: Write the polarization basis vectors $\hat{\boldsymbol{m}}$ and $\hat{\boldsymbol{n}}$ perpendicular to $\hat{\boldsymbol{k}}$, then write e^{+}_{ij} and $e^{\times}_{ij}$.

Step 2: Write F^{+}_{a} and $F^{\times}_{a}$ for pulsar 1 (along $\hat{z}$).

Step 3: Write F^{+}_{b} and $F^{\times}_{b}$ for pulsar 2 (at angle ζ).

Step 4: Form $\sum_A F^A_a F^A_b$ and set up the $d\theta\, d\phi$ integral.

Result: The Hellings-Downs Curve

Final Result (fill in after board derivation)

$$\chi(\zeta) = \frac{1}{2} - \frac{1}{4}\left(\frac{1-\cos\zeta}{2}\right) + \frac{3}{2}\left(\frac{1-\cos\zeta}{2}\right)\ln\left(\frac{1-\cos\zeta}{2}\right)$$

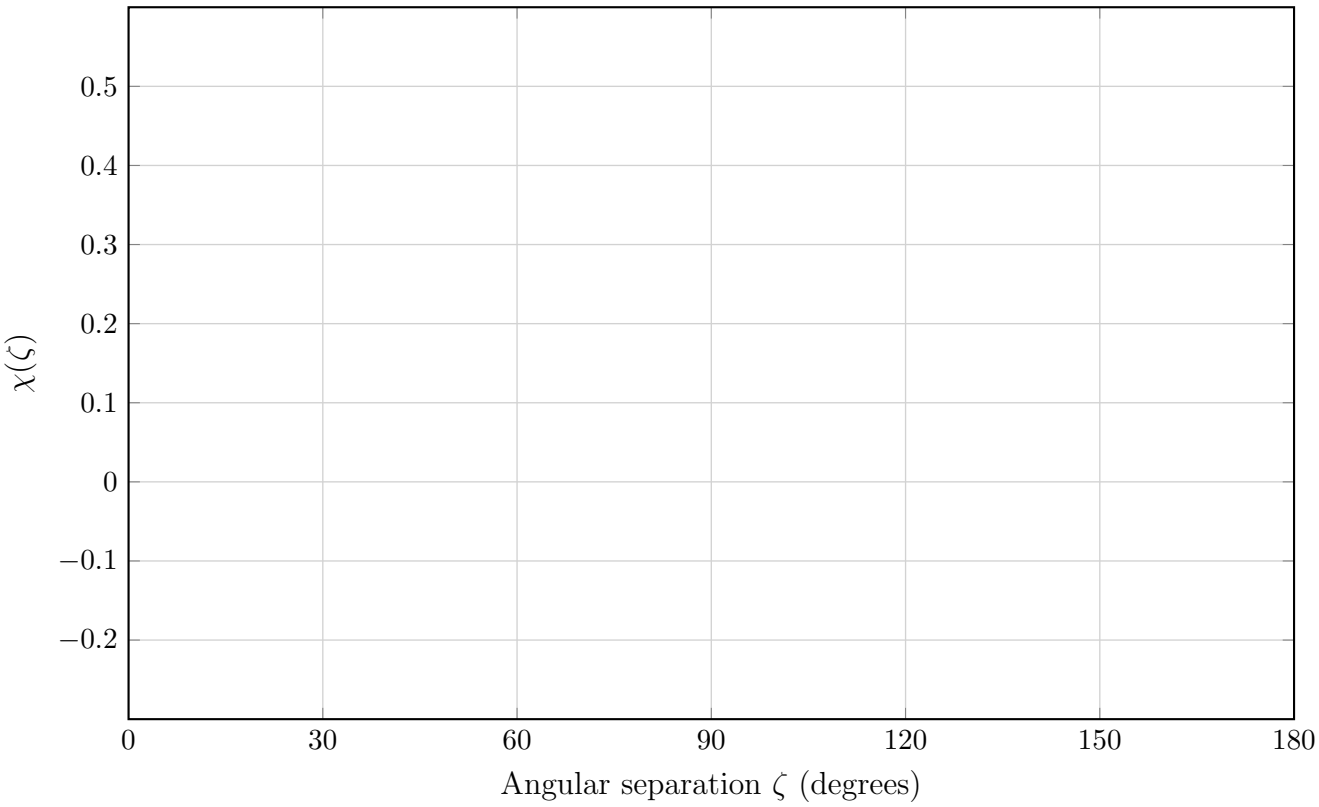
Normalisation: $\chi(0) =$ _____ (self-correlation; explain physically):

_____

Interpret each key feature:

- $\chi(0) = 1/2$: _____
- $\chi \approx 0$ near 90: _____
- Negative χ near 120: _____
- ln term: _____

Plot the Hellings-Downs curve on the axes below:



Why is the HD curve the "smoking gun" for an SGWB in GR?

- Different GW polarisations (scalar, vector) predicted by alternative theories of gravity give *different* ORFs. Only GR tensor modes give the HD curve.
- Clock errors → _____ correlation; solar system ephemeris errors → _____ correlation. *These are systematics to reject.*
- Detection strategy: compare measured pulsar-pair correlations to HD prediction. Statistical significance of HD-consistent correlations = evidence for SGWB.
- _____
- _____

4 PTA Instrumentation: From Telescope to Science

Signal Chain Label the signal processing chain below:



Key Instrumentation Concepts

1. ToA uncertainty (radiometer-noise limit):

$$\sigma_{\mathrm{ToA}} \approx \frac{W^{1/2}}{S\,(\Delta\nu)^{1/2}} \propto \frac{T_{\mathrm{sys}}}{G\,S\sqrt{\Delta\nu\,t_{\mathrm{int}}}}$$

What does this tell us about the telescope/backend design tradeoffs?

_____

2. Coherent dedispersion:

A standard filterbank removes _____ smearing *within* channels.
Coherent dedispersion operates on the _____ stream, removing all dispersion to the _____ level.
Why is this important for μs-precision timing? _____

3. Wideband timing:

Modern receivers (0.7–4 GHz) enable simultaneous _____ measurement and ToA precision. Name one example receiver/backend: _____.

4. Clock referencing and systematics:

Clock errors ⇒ _____ correlation in residuals (monopole).

Solar system ephemeris errors ⇒ _____ correlation (dipole).

SGWB ⇒ _____ correlation (quadrupole). ✓

Complete the table of major PTAs and their primary telescopes:

PTA	Primary Telescope(s)	Data span	No. pulsars
NANOGrav			
EPTA			
PPTA			
CPTA			
MPTA			

4 Big Ideas: Synthesis

Lecture 1 Summary (complete as instructor reviews):

1. A PTA is a _____-scale GW detector: pulsars → _____, residuals → _____, HD curve → _____.
2. GWs from a BBH inspiral: $f_{\rm gw} = 2f_{\rm orb}$; characteristic strain $h_c \propto f$ —— for a cosmic SMBHB population.
3. HD curve: $\chi(\zeta) = \frac{1}{2} - \frac{1}{4}\left(\frac{1-\cos\zeta}{2}\right) + \frac{3}{2}\left(\frac{1-\cos\zeta}{2}\right)\ln\left(\frac{1-\cos\zeta}{2}\right)$ — uniquely predicted by _____ for tensor GW polarisations.
4. Timing precision $\sigma_{\rm ToA} \propto T_{\rm sys}/(G\sqrt{\Delta\nu\, t_{\rm int}})$ directly controls PTA sensitivity to h_c.
5. As of 2023–2025: _____–_____σ evidence for SGWB from NANOGrav/EPTA/PPTA/CPTA; not yet a definitive 5σ detection.

Three Big Questions (fill in answers as instructor provides them):

1. **What is the origin of the detected GW background?**
2. **Why is $A_{\rm gw}$ higher than expected, and what does it imply about SMBHB populations?**
3. **What are the next observational milestones?**

Preview — Lecture 2 (Jupyter Notebook Demo):

- Load real NANOGrav pulsar timing data (par/tim format) with `enterprise`
- Bayesian noise analysis: recover $A_{\rm RN}$, $\gamma_{\rm RN}$ for individual pulsars
- Build PTA sensitivity curves and HD correlation with `hasasia`
- Search for the SGWB; compute the optimal statistic
- Use `holodeck` to connect $A_{\rm gw}$ to SMBHB population constraints

Suggested Reading:

- Jenet & Romano (2015) arXiv:1412.1142 — HD curve via sound/EM analogy
- NANOGrav 15-yr GWB paper (Agazie et al. 2023, ApJL 951, L8)
- Buonanno (2007) arXiv:0709.4682 — GW from binary black holes
- Phinney (2001) arXiv:astro-ph/0108028 — GW background from cosmological populations
- Hazboun et al. (2019) *hasasia* JOSS paper
- NANOGrav public tutorials